

CIP-B101 Basic Computer Skills for Digital Forensics

LAB 2B: Advanced Linux Commands, Standard Streams and Controlled Evidence Transfer

Student Practical Lab Manual

Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Code	Lab 2B
Lab Theme	Advanced Linux command-line use for forensic evidence handling, stream redirection, and safe file acquisition
Recommended Duration	3-4 hours practical work + report preparation
Submission Format	PDF report + zipped evidence folder
Prepared For	ICDFA trainees

Source material: Advanced Linux Commands - Remote File/Disk Acquisition slide deck provided for CIP-B101.

Important Student Preparation Notice

Before starting this lab, trainees must carefully go through the course slide material on Advanced Linux Commands and Remote File/Disk Acquisition. The slide is a walkthrough and explains the concepts behind standard input, standard output, standard error, file descriptors, netcat communication, and remote evidence transfer. Do not start the practical tasks until you understand the concepts and the safety boundaries.

1. Lab Scenario

You are a junior digital forensic trainee assigned to practice safe command-line evidence handling between two authorized lab machines. Your task is to understand how Linux treats input/output devices as files, how command output can be redirected into evidence logs, and how a file can be transferred across a controlled network without altering its content. You will document every action and verify file integrity using cryptographic hashes.

This lab intentionally avoids unauthorized access. You will not perform real backdoor deployment or uncontrolled shell access. Any bind-shell or reverse-shell concept in the lecture material is treated as theory only, useful for understanding attacker behavior and forensic risk. The practical activity is limited to authorized file transfer and small instructor-approved evidence images.

Safety and Authorization Rule

Only use machines, virtual machines, files and networks that you own or that your instructor has authorized. Do not run shell execution options, backdoor commands, or reverse shell payloads against any real system. In this lab, netcat/ncat is used only as a controlled file transfer and evidence acquisition tool.

2. Learning Objectives

- Explain standard input, standard output and standard error using file descriptors 0, 1 and 2.
- Redirect command output and errors into evidence files for documentation.
- Use safe netcat/ncat communication to transfer a file between two authorized lab machines.
- Create and verify hashes before and after evidence transfer.
- Create a small test evidence image using dd and transfer it safely across a lab network.
- Document the process using screenshots, command logs and a short forensic interpretation report.
- Explain why remote acquisition must be controlled, authorized and repeatable in digital forensics.

3. Required Lab Setup

Machine 1	Kali Linux or Ubuntu Linux VM
Machine 2	Windows 10/11 VM or host computer used only with permission
Network	Both machines must be on the same isolated lab network or internal VM network
Linux Tools	bash, cat, echo, sha256sum, md5sum, dd, nc/netcat or ncat, ip command
Windows Tools	Command Prompt or PowerShell, certutil, ncat from Nmap, text editor
Permissions	Administrator permission may be needed for installation, but evidence transfer tasks do not require attacking a system
Storage	At least 500 MB free space for test files and screenshots

4. Relevant URLs Extracted from the Slide

The following URLs were referenced in the slide material and are included here so students can access the required tools and supporting readings:

Resource	URL
Nmap/Ncat download	https://nmap.org/download
Windows dd utility	http://www.chrysocome.net/downloads/dd-0.5.zip
dd block-size testing script	https://raw.githubusercontent.com/tdg5/blog/master/includes/scripts/dd_obs_test.sh
Shell scripting reference	http://www.learnlinux.org.za/courses/build/shell-scripting/ch01s04.html
Standard streams reference	https://en.wikipedia.org/wiki/Standard_streams
Shell redirection explanation	https://www.briantorti.com/understanding-shell-script-idiom-redirect/
Netcat command reference	https://www.computerhope.com/unix/nc.htm
SANS Netcat cheat sheet	https://www.sans.org/security-resources/sec560/netcat_cheat_sheet_v1.pdf
Reverse shell concept reference for theory only	https://stackoverflow.com/questions/35271850/what-is-a-reverse-shell

Note: The reverse shell resource is included only to help students understand terminology and forensic risk. Students must not execute reverse shell commands in this lab.

5. Submission Requirements

- A PDF report named: CIP-B101_Lab2B_Firstname_Lastname.pdf
- A zipped evidence folder named: CIP-B101_Lab2B_Firstname_Lastname_Evidence.zip
- Screenshots showing commands, outputs, IP addresses, file hashes, transfer results and folder structure.
- All generated evidence files, command logs and hash files.
- Short answers to the reflection questions at the end of the lab.

6. Evidence Folder Structure

Create the following folder structure at the start of the lab. Replace Firstname_Lastname with your real name.

```
mkdir -p ~/CIP-
B101_Lab2B_Firstname_Lastname/{screenshots,logs,evidence,received,hashes,report_notes}
cd ~/CIP-B101_Lab2B_Firstname_Lastname
pwd
ls -la
```

Screenshot required: Terminal showing the created lab folder structure.

7. Part A - Standard Input, Output and Error

In Linux and Unix-like systems, standard input, standard output and standard error are represented using file descriptors. This is important in forensics because command output can be preserved as evidence logs.

stdin	File descriptor 0 - default input source, usually the keyboard or redirected input file
stdout	File descriptor 1 - normal command output, usually displayed on the terminal
stderr	File descriptor 2 - error messages, usually displayed on the terminal

Task A1 - Create a Test Evidence Text File

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/evidence
echo "This is an authorized forensic evidence transfer test for Firstname Lastname" >
evidence_note.txt
cat evidence_note.txt
ls -l evidence_note.txt
```

Screenshot required: evidence_note.txt content and file listing.

Task A2 - Redirect Standard Output

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname
ls -la evidence > logs/evidence_listing_stdout.txt
cat logs/evidence_listing_stdout.txt
```

Explain in your report what the > symbol did and why this is useful for evidence documentation.

Task A3 - Redirect Standard Error

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname
cat evidence/file_that_does_not_exist.txt 2> logs/error_stderr.txt
cat logs/error_stderr.txt
```

Screenshot required: error message saved inside error_stderr.txt.

Task A4 - Redirect stdout and stderr Together

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname
(ls evidence && cat evidence/file_that_does_not_exist.txt) > logs/combined_output.txt 2>&1
cat logs/combined_output.txt
```

In your report, explain the meaning of 2>&1.

8. Part B - Baseline System and Network Documentation

Before transferring evidence, document both lab machines. This is similar to recording acquisition environment details in a forensic notebook.

Task B1 - Record Linux Network Details

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname
hostname | tee logs/linux_hostname.txt
ip address | tee logs/linux_ip_address.txt
ip route | tee logs/linux_ip_route.txt
which nc || which ncat || echo "netcat or ncat not found" | tee logs/netcat_check.txt
```

Screenshot required: IP address of the Linux machine.

Task B2 - Record Windows Network Details

On Windows Command Prompt or PowerShell, run:

```
hostname
ipconfig
where ncat
certutil -?
```

Screenshot required: IP address of the Windows machine and confirmation that ncat is available.

Task B3 - Connectivity Test

From Linux, ping the Windows machine using its IP address. Replace WINDOWS_IP with the actual IP address.

```
ping -c 4 WINDOWS_IP | tee logs/ping_windows.txt
```

From Windows, ping the Linux machine using its IP address. Replace LINUX_IP with the actual IP address.

```
ping LINUX_IP
```

Screenshot required: successful ping test from at least one side. If ping is blocked, explain the likely firewall reason and proceed only if file transfer works.

9. Part C - Hash the Evidence Before Transfer

A cryptographic hash helps prove that the transferred file is identical to the source file. You must record the hash before and after transfer.

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/evidence
sha256sum evidence_note.txt | tee ../hashes/source_evidence_note_sha256.txt
md5sum evidence_note.txt | tee ../hashes/source_evidence_note_md5.txt
cat ../hashes/source_evidence_note_sha256.txt
```

Screenshot required: SHA256 and MD5 hash values of the source evidence file.

10. Part D - Controlled Evidence File Transfer from Linux to Windows

This task uses netcat/ncat only to transfer a file. Do not use shell execution options such as -e or --exec. This is not a backdoor exercise.

Task D1 - Start Listener on Windows

On Windows, open Command Prompt in the folder where you want to receive the file. Example:

```
mkdir C:\CIP-B101_Lab2B_Received
cd C:\CIP-B101_Lab2B_Received
ncat -l -p 9090 > received_evidence_note.txt
```

Leave the Windows terminal waiting. It will finish after Linux sends the file.

Task D2 - Send File from Linux

On Linux, replace WINDOWS_IP with the Windows IP address.

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/evidence
nc WINDOWS_IP 9090 < evidence_note.txt
```

If your Linux uses ncat instead of nc, use:

```
ncat WINDOWS_IP 9090 < evidence_note.txt
```

Task D3 - Verify the Received File on Windows

On Windows, verify the file hash:

```
cd C:\CIP-B101_Lab2B_Received
certutil -hashfile received_evidence_note.txt SHA256
certutil -hashfile received_evidence_note.txt MD5
type received_evidence_note.txt
```

Screenshot required: received file content and hash value. Compare it with the Linux source hash.

11. Part E - Controlled Evidence File Transfer from Windows to Linux

This part reverses the transfer direction. Students should create a file in Windows, then acquire it from Linux through a controlled listener. Again, no shell access is allowed.

Task E1 - Create Evidence File on Windows

On Windows, create the evidence file. Replace FirstnameLastname with your real name.

```
cd C:\CIP-B101_Lab2B_Received
echo hello Firstname Lastname! > FirstnameLastnameEvidence.txt
type FirstnameLastnameEvidence.txt
certutil -hashfile FirstnameLastnameEvidence.txt SHA256
```

Task E2 - Start Listener on Linux

On Linux, prepare to receive the Windows file.

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/received
nc -l -p 9091 > FirstnameLastnameEvidence_received.txt
```

If your version of nc requires a different syntax, try:

```
ncat -l -p 9091 > FirstnameLastnameEvidence_received.txt
```

Task E3 - Send from Windows

On Windows, replace LINUX_IP with the Linux IP address.

```
ncat LINUX_IP 9091 < FirstnameLastnameEvidence.txt
```

Task E4 - Verify on Linux

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/received
cat FirstnameLastnameEvidence_received.txt
sha256sum FirstnameLastnameEvidence_received.txt | tee ../hashes/received_windows_file_sha256.txt
```

Screenshot required: Linux terminal showing received file content and SHA256 hash.

12. Part F - Small Test Image Transfer Using dd

The slide discusses remote disk acquisition. For student safety, this lab uses a small instructor-approved test image instead of imaging a real system disk. This teaches the same integrity and transfer principles without risking damage or unauthorized acquisition.

Task F1 - Create a Small Test Evidence Image

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/evidence
dd if=/dev/zero of=test_evidence_image.dd bs=1M count=5 status=progress
echo "ICDFA test evidence marker - Firstname Lastname" | dd of=test_evidence_image.dd bs=1
seek=1024 conv=notrunc
ls -lh test_evidence_image.dd
sha256sum test_evidence_image.dd | tee ../hashes/source_test_image_sha256.txt
```

Screenshot required: created image size and SHA256 hash.

Task F2 - Transfer the Test Image to Windows

On Windows, start listener:

```
cd C:\CIP-B101_Lab2B_Received
ncat -l -p 9092 > test_evidence_image_received.dd
```

On Linux, send the image:

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/evidence
nc WINDOWS_IP 9092 < test_evidence_image.dd
```

Task F3 - Verify Image Hash on Windows

```
cd C:\CIP-B101_Lab2B_Received
certutil -hashfile test_evidence_image_received.dd SHA256
```

The hash must match the Linux source hash. If it does not match, the transfer is not forensically reliable and must be repeated.

13. Part G - Optional dd Block Size Test

This optional section introduces the idea that block size affects dd performance. Run this only on Linux and only against harmless test files.

```
cd ~/CIP-B101_Lab2B_Firstname_Lastname/logs
wget https://raw.githubusercontent.com/tdg5/blog/master/_includes/scripts/dd_obs_test.sh
chmod +x dd_obs_test.sh
./dd_obs_test.sh | tee dd_block_size_test_result.txt
```

Screenshot required if attempted: output showing the tested block sizes and recommended/fastest result.

14. Part H - Theory-Only Security Reflection

The lecture material introduces bind shell and reverse shell concepts because they are important in cybersecurity and forensic investigations. In this student lab, you must not execute any bind shell or reverse shell. Instead, answer the questions below based on your understanding of the slide material.

1. What is the difference between standard output and standard error?
2. Why is command redirection useful during digital forensic documentation?
3. Why must a forensic examiner calculate the hash of a file before and after transfer?
4. Why is it unsafe and unethical to use remote shell access on a system without authorization?
5. What is the difference between a bind shell and a reverse shell at a conceptual level?
6. Why might a firewall block one type of connection but allow another?
7. Why did this lab use a small test evidence image instead of imaging a real disk?

15. Screenshots Checklist

- ☐ Lab folder structure created successfully.
- ☐ evidence_note.txt created and displayed.
- ☐ stdout redirected to evidence_listing_stdout.txt.
- ☐ stderr redirected to error_stderr.txt.
- ☐ combined stdout/stderr saved to combined_output.txt.
- ☐ Linux IP address and routing information captured.
- ☐ Windows IP address captured.
- ☐ Successful connectivity test or explanation of blocked ping.
- ☐ Source evidence file hash captured before transfer.
- ☐ Linux-to-Windows transfer completed and verified.
- ☐ Windows-to-Linux transfer completed and verified.
- ☐ Small dd image created and hashed.
- ☐ Small dd image transferred and verified.
- ☐ Final evidence folder compressed for submission.

16. Final Report Structure

8. Cover page: student name, course code, lab title and date.
9. Lab objective: what the lab was designed to achieve.
10. Lab environment: OS versions, VM names, IP addresses, tools used.
11. Procedure: clear step-by-step explanation of what you did.
12. Evidence integrity: source hash, destination hash and comparison result.
13. Screenshots: properly labelled and explained.
14. Findings: what you proved from the transfer and hash verification.
15. Reflection answers: answer all theory questions.
16. Conclusion: what you learned and how it applies to digital forensics.

17. Final Evidence Packaging

On Linux, compress your evidence folder after completing the report. Replace Firstname_Lastname with your real name.

```
cd ~
tar -czvf CIP-B101_Lab2B_Firstname_Lastname_Evidence.tar.gz CIP-B101_Lab2B_Firstname_Lastname
ls -lh CIP-B101_Lab2B_Firstname_Lastname_Evidence.tar.gz
```

18. Marking Rubric - 100 Marks

Assessment Area	Marks	Instructor Comment
Preparation and lab environment documentation	10	

Correct understanding and use of stdin, stdout and stderr	15	
Accurate command logging and redirection evidence	10	
Successful Linux-to-Windows file transfer	10	
Successful Windows-to-Linux file transfer	10	
Hash verification before and after transfer	15	
Small dd image creation, transfer and verification	10	
Screenshots and evidence organization	10	
Reflection answers and forensic interpretation	5	
Professional report formatting and submission quality	5	

19. Common Mistakes to Avoid

- Do not use shell execution options with netcat/ncat in this lab.
- Do not image your full system disk or another person's device.
- Do not skip hash verification.
- Do not submit screenshots without explaining what each screenshot proves.
- Do not use different filenames from the required submission naming format.
- Do not run commands against public IP addresses or unauthorized machines.